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Bachelor Of Technology in Software Technology

Anuraa Cottages – Smart Online Cabana Booking and Management System

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Preamble

This project proposal presents a plan to design and develop a **Smart Online Cabana Booking and Management System** for *Anuraa Cottages*, located near Ella Rock, Sri Lanka. The system aims to digitalize cabana booking operations, improve workflow efficiency, and enhance guest satisfaction by integrating technology into the management process.

Currently, **around 20–30 reservations per month** are handled manually through phone calls, WhatsApp messages, and social media inquiries, leading to an **average response delay of about 10–15 minutes per inquiry**. This manual process often results in booking overlaps, delayed confirmations, and difficulties in tracking seasonal occupancy.

The proposed system targets **three main user groups** — administrators, staff members, and customers — and will introduce several key advantages such as **real-time synchronization**, **automated booking management**, and a **responsive web design** accessible from both desktop and mobile devices. These enhancements will streamline daily operations, reduce human errors, and provide a modern and efficient booking experience for guests.

1. Introduction

Anuraa Cottages is a small-scale tourism business that offers cabana-style lodging facilities with scenic mountain views. The location attracts both local and international travelers who often prefer digital and fast booking experiences.

Currently, cabana reservations are managed manually through phone calls, WhatsApp messages, and social media inquiries. This manual approach causes delays, booking overlaps, and miscommunication between staff and customers.

To address these challenges, a **Smart Online Cabana Booking and Management System** will be developed to automate the booking process, manage cabana availability, and simplify payment handling through an integrated web-based platform.

2. Background

Tourism in Ella has significantly increased over the past few years. Guests now expect to find real-time accommodation availability online and complete bookings with just a few clicks. However, *Anuraa Cottages* still maintains manual booking logs and communication channels, which limit scalability and customer convenience.

Although third-party platforms such as **Booking.com** and **Airbnb** provide online booking services, they are not fully optimized for the **specific operational needs** of Anuraa Cottages. These external systems charge commissions, limit customization options, and restrict direct communication with customers. They also lack integration with internal processes such as **staff task management**, **cabana maintenance tracking**, and **custom seasonal pricing**.

The proposed solution will therefore introduce a **centralized and fully customizable online platform** that enables:

- Real-time cabana availability checking
- Secure online booking and payment
- Administrative control over reservations and pricing
- Automated report generation
- Direct communication between guests and management

By implementing this tailor-made system, Anuraa Cottages can strengthen its brand identity, enhance customer trust, and operate more efficiently compared to relying on generic third-party platforms.

3. Problem Description / Gap Analysis

Existing Problems:

1. Manual booking process prone to human errors.
2. Lack of online visibility and payment integration.
3. Difficulty tracking seasonal pricing and occupancy rates.
4. No structured feedback collection from customers.

Desired Outcome:
A web-based system that automates bookings, provides secure online payments, generates management reports, and maintains customer records efficiently.

Gap Analysis:

The existing manual process at *Anuraa Cottages* creates a clear service gap between customer expectations and actual performance. The following table compares the **current manual system** with the **proposed Smart Online Cabana Booking and Management System** across key functional and non-functional areas:

Aspect	Current Manual System (Anuraa Cottages)	Proposed Smart Online System
Booking Management	Managed via phone calls, WhatsApp, and social media messages — prone to overlaps and missed reservations.	Fully automated online booking with real-time availability updates and instant confirmations.

Customer Communication	Manual responses cause delays (average 10–15 minutes).	Instant notifications and automated email confirmations improve customer response time.
Data Management	Paper-based or spreadsheet records, difficult to track and analyze.	Centralized database with searchable records and reporting tools.
Payment Handling	Only cash or manual bank transfers.	Integrated secure online payments via **PayHere API**.
Report Generation	Manual compilation of monthly records.	Automated report generation with customizable filters.
Cost Efficiency	Requires more staff time and manual effort; potential human errors increase costs.	Reduces labor costs and eliminates commission fees from third-party platforms like Booking.com.
Security	No formal security measures; data risk from loss or misplacement.	Role-based authentication, encrypted data handling, and secure login mechanisms.
Redundancy / Backup	No regular backups; data loss possible during system failures.	Automated daily backups and recovery mechanisms ensure data redundancy.
Scalability	Difficult to expand or integrate new features.	Scalable design using Laravel, Vue.js, and MySQL for future growth.
User Experience (UX)	Non-interactive, no visual interface for customers.	Responsive and user-friendly design accessible on both mobile and desktop devices.

Summary:
By adopting this system, *Anuraa Cottages* will bridge the gap between traditional manual operations and modern digital expectations. The proposed platform ensures higher efficiency, security, and customer satisfaction while maintaining lower operational costs.

4. Aim and Objectives

4.1 Aim

To develop a **web-based Smart Cabana Booking and Management System** that automates the reservation process, integrates secure online payments, and enhances the overall operational efficiency of *Anuraa Cottages* by delivering faster, more reliable, and data-driven service.

4.2 Objectives

Objective	**Performance / Measurement Criteria**
1. Build a responsive online booking interface for customers.	System should load the main booking page within **3 seconds** on standard internet connections.
2. Automate the cabana reservation and confirmation process.	Each booking request should be processed and confirmed within **2 seconds** per request.
3. Integrate a secure online payment gateway (PayHere).	100% of transactions should use SSL encryption with **zero payment data leaks**.
4. Develop an admin dashboard for staff to manage bookings, pricing, and reports.	Dashboard response time should not exceed **2 seconds per operation**.
5. Generate automated reports and analytical summaries.	Reports for a one-month period should be generated within **5 seconds**.
6. Implement feedback and rating features for service improvement.	At least **70%** of monthly users should provide feedback within three months of deployment.
7. Ensure system reliability and data security.	Maintain **99% uptime** and perform **daily automated backups**.
8. Enhance overall efficiency and accuracy of booking operations.	Reduce manual workload by **at least 50%** within the first three months of implementation.

5. Methodology

The project follows the **Waterfall Model**, which is suitable for projects with **clearly defined requirements** and a **limited scope**. Each phase of the model will be completed in a sequential order to ensure systematic progress and proper documentation.

Why Waterfall Model

The **Waterfall approach** is selected because it provides a clear and structured workflow with well-defined deliverables at each stage. Since this project involves specific requirements and limited team members, the sequential process ensures better control, traceability, and reduced rework. This model is particularly effective for small-scale projects where requirements are stable and easy to define from the beginning.

Project Phases

1. **Requirement Analysis** – Collect detailed system requirements from management and staff through interviews and observation.
2. **System Design** – Develop system architecture, database design, and user interface wireframes using tools such as **Draw.io** and **Figma**.
3. **Development** – Implement the backend using **Laravel 11** and the frontend using **Vue.js**, connected via RESTful APIs.
4. **Testing** – Perform functional and performance testing using:
    - **Postman** for API validation
    - **PHPUnit** for backend unit testing
    - **Manual user acceptance testing (UAT)** to verify overall system usability and stability
5. **Deployment** – Deploy the system on a **cPanel hosting environment** (or **Laravel Forge / AWS EC2**) with **MySQL** managed through **phpMyAdmin** for database administration.
6. **Maintenance** – Regularly monitor the system, apply updates, fix bugs, and add new features based on user feedback.

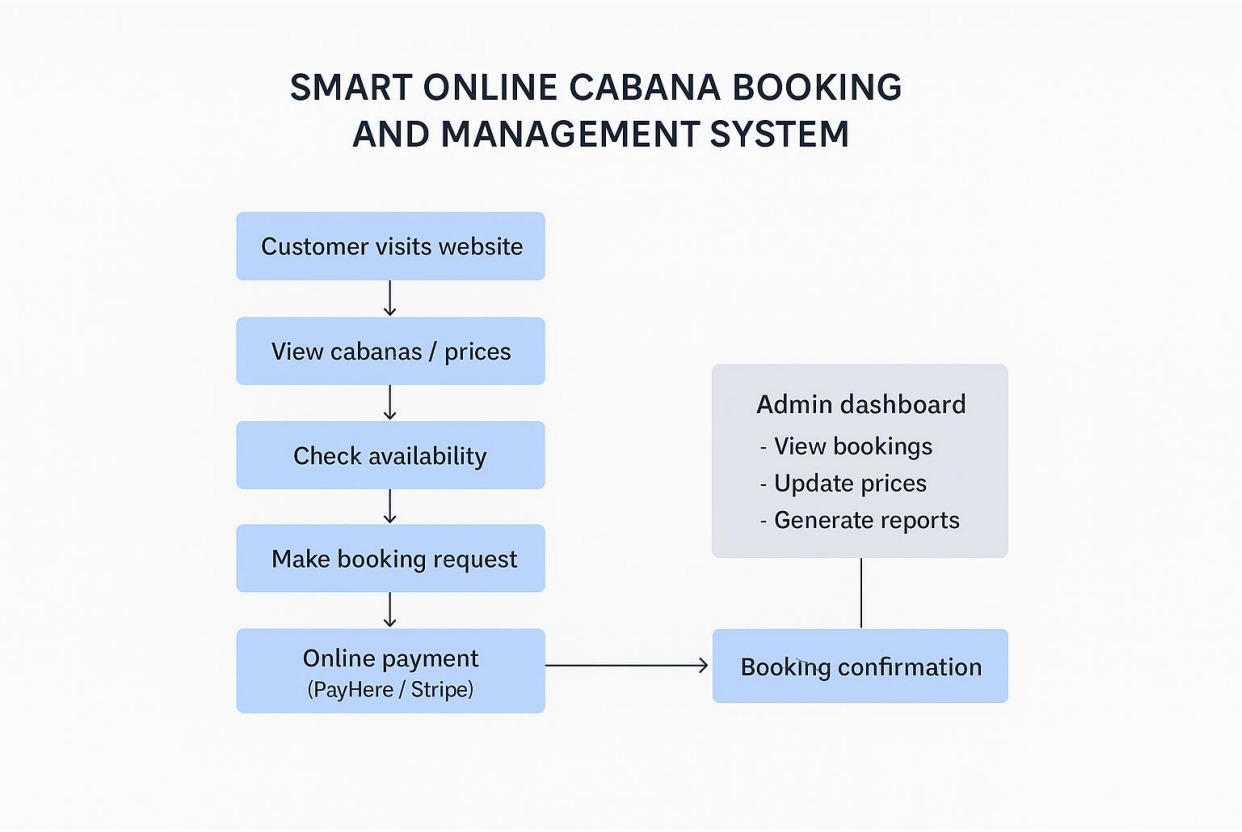
Process Flow

Phase	Key Tasks	Deliverables
Requirement Analysis	Interviews, data collection	Requirement documentation
System Design	UI mockups, ERD, architecture design	Design specifications
Development	Backend + Frontend coding	Functional system prototype
Testing	API & unit testing, UAT	Test reports and bug fixes
Deployment	Host on cPanel / Laravel Forge / AWS	Live, accessible website
Maintenance	Updates and enhancements	Improved system versions

Tools & Technologies Used

- **Frontend:** Vue.js
- **Backend:** Laravel 11
- **Database:** MySQL
- **Testing Tools:** Postman, PHPUnit
- **Deployment Platforms:** cPanel / Laravel Forge / AWS
- **Design Tools:** Figma, Draw.io
- **Version Control:** GitHub

5.1 Process Flow Diagram



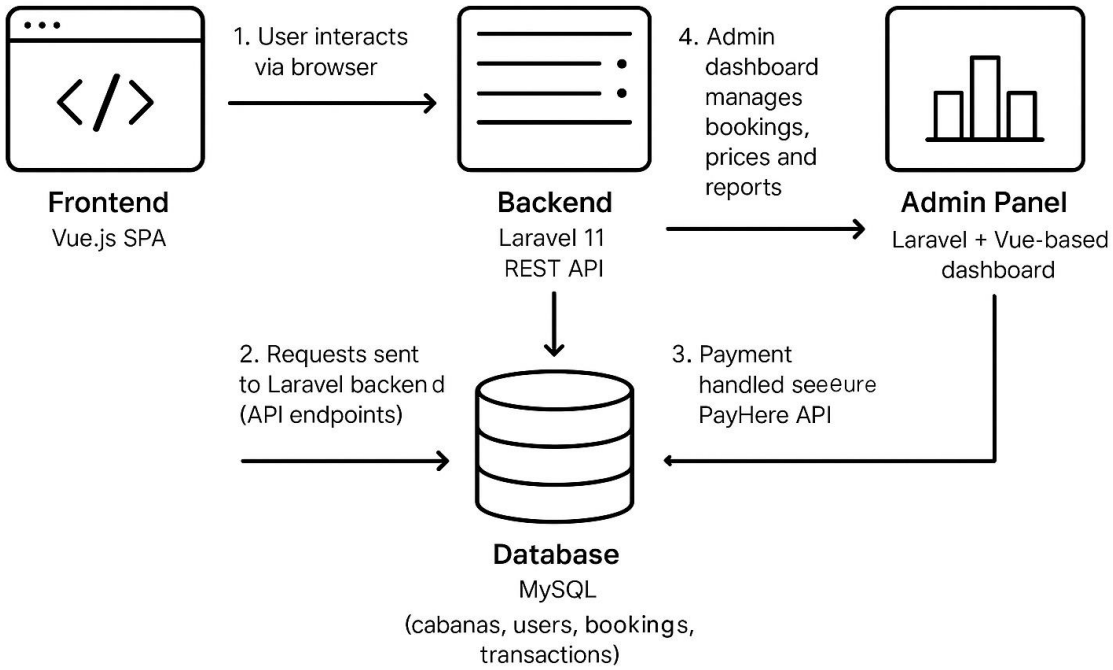
5.2 Work Breakdown Structure (WBS)

Phase	**Task**	**Deliverables**
1	Requirement Analysis	Requirement documentation
2	System Design	UI wireframes, ER diagram
3	Development	Functional booking system
4	Testing	Bug reports, test summary
5	Deployment	Hosted system
6	Maintenance	Updated version with feedback

5.3 Project Timeline

Week	Activity	Milestone
1–2	Requirement Analysis	Requirements approved
3–5	Design Phase	Completed system design
6–10	Development	Working prototype
11	Testing	System verified
12	Deployment	Live website launched

5.4 System Architecture Diagram



The architecture diagram should include:

Components:

- Frontend: Vue.js SPA running in the customer browser
- Backend: Laravel 11 REST API hosted on a web server
- Database: MySQL (cabanas, users, bookings, transactions)
- Payment Gateway: PayHere API for secure transactions
- Admin Panel: Laravel + Vue-based dashboard accessible to staff

Data Flow Overview:

1. User interacts via browser (Vue.js interface)
2. Requests sent to Laravel backend (API endpoints)
3. Backend retrieves/stores data in MySQL
4. Payment handled securely through PayHere API
5. Admin dashboard manages bookings, prices, and reports

6. Expected Deliverables

The following outputs are expected upon completion of the project:

1. Fully functional web-based cabana booking system.
2. Admin control panel for staff management.
3. Integrated online payment gateway.
4. Automated report generation module.
5. System documentation (technical + user manual).
6. Final project presentation and demonstration.

7. Risk Analysis and Mitigation

A successful software project must anticipate potential risks and establish strategies to minimize their impact. The following table outlines the major risks identified for the Smart Online Cabana Booking and Management System and the corresponding mitigation strategies.

Risk	**Impact Level**	**Mitigation Strategy**
Internet Downtime	Medium	Implement offline caching and auto-sync mechanisms to temporarily store booking data and update the database once the connection is restored.
Payment Gateway Failure	High	Integrate multiple payment options, including **QR-based payments** and **manual confirmation fallback**. Ensure **SSL certification** for secure data transmission.

Unauthorized Access / Data Breach	High	Apply **dual authentication** (2FA) for admin and staff logins. Encrypt all sensitive customer information using **AES-256 encryption** standards.
Customer Data Security	High	Store passwords using **bcrypt hashing**. Apply HTTPS with **SSL certificates** to ensure end-to-end encrypted communication between client and server.
System Downtime or Data Loss	High	Enable **automated daily backups** on both the web server and remote storage. Conduct periodic restore tests.
Development Delays	Medium	Allocate buffer time in the schedule and use milestone-based tracking to monitor progress.
Software Bugs After Deployment	Medium	Conduct regression testing and schedule regular maintenance updates to ensure continuous improvement.
User Mismanagement / Improper Training	Low	Provide documentation and user manuals. Conduct a training session for staff before system rollout.

Additional Security Measures:

- Implementation of **SSL Certificates** to secure all data exchanges.
- **Dual Authentication (2FA)** for critical user roles.
- **QR Payment Integration** through PayHere to ensure fast and verified transactions.
- Continuous **encryption of customer and payment data** both in storage and transmission.
- Regular **security audits and penetration testing** to identify vulnerabilities early.

8. References

- Laravel Official Docs – https://laravel.com/docs
- Vue.js Official Docs – https://vuejs.org/
- PayHere API Documentation – https://www.payhere.lk
- MySQL Reference Manual – https://dev.mysql.com/doc/
- Welling, L. & Thomson, L. (2017). *PHP and MySQL Web Development.* Addison-Wesley.

Appendices

Appendix A – Process Flow Diagram
Appendix B – System Architecture Diagram